TERTIARY INFOTECH ACADEMY

LAB 08
Diagnose IAM Access and Protect Data with KMS

AWS Certified SysOps Administrator Associate Training
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Scenario

A learner workload is denied access to an encrypted object after a policy change, while an earlier role remains broader than required.

Objective

Apply K1, K3, A1 and A7 by explaining an access decision, narrowing permissions and verifying encryption and key-policy behaviour.

Deliverable

A policy-evaluation worksheet, corrected least-privilege policy, KMS key-policy review and encrypted-object evidence.

Services and tools

IAM · Access Analyzer · Policy Simulator · KMS · S3 · CloudTrail

Safety and cost boundary

- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Read `ACCESS-SCENARIO.md`; identify principal, action, resource, conditions, boundary and possible explicit denies.

```
open ACCESS-SCENARIO.md
```

Evidence note: _____

Step 2 — Validate the supplied identity, bucket and KMS policy JSON before reasoning about the decision.

```
for f in identity-policy.json bucket-policy.json kms-key-policy.json; do python3 -m json.tool
"$f" >/dev/null; done
```

Evidence note: _____

Step 3 — Use the IAM policy simulator or trainer-provided equivalent to test GetObject on the exact learner object ARN.

```
aws iam simulate-principal-policy --policy-source-arn REPLACE_ROLE_ARN --action-names
s3:GetObject kms:Decrypt --resource-arns REPLACE_OBJECT_ARN REPLACE_KEY_ARN | tee
evidence/simulation.json
```

Evidence note: _____

Step 4 — Inspect CloudTrail for the denied request and record error code, principal, resource and UTC time.

```
aws cloudtrail lookup-events --lookup-attributes
AttributeKey=EventName,AttributeValue=GetObject --max-results 20 | tee evidence/cloudtrail-
deny.json
```

Evidence note: _____

Step 5 — Complete `ACCESS-DECISION.md` in evaluation order; distinguish the S3 allow from the KMS decrypt requirement.

```
open ACCESS-DECISION.md
```

Evidence note: _____

Step 6 — Edit `identity-policy.json` to the exact bucket prefix and key; retain no wildcard action.

```
open identity-policy.json
```

Evidence note: _____

Step 7 — Upload and retrieve a small encrypted learner-owned object, then verify encryption metadata.

```
aws s3api head-object --bucket REPLACE_BUCKET --key lab08/evidence.txt | tee evidence/head-
object.json
```

Evidence note: _____

Step 8 — Run Access Analyzer validation on the policies and resolve ERROR findings.

```
aws accessanalyzer validate-policy --policy-document file://identity-policy.json --policy-
type IDENTITY_POLICY | tee evidence/policy-validation.json
```

Evidence note: _____

Step 9 — Capture policy simulation and encrypted-object evidence as JPEG screenshots; run validation.

```
open EXPECTED-ARCHITECTURE.jpg
bash validate.sh
```

Evidence note: _____

Step 10 — Remove the learner test object and any temporary role/policy; do not schedule KMS key deletion without trainer approval.

```
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

The decision path explains the deny, corrected policy uses exact actions/resources, encrypted-object metadata is verified, Access Analyzer reports no ERROR, and validation emits LAB 08 PASS.

Acceptance criteria

- No wildcard action
- Resource scope is exact

- KMS and S3 permissions both considered
- Explicit deny precedence explained
- Temporary access is removed

Evidence checklist

□ Policy simulation

□ CloudTrail denied event

□ policy validation

□ encrypted object metadata

□ completed ACCESS-DECISION.md

□ JPEG evidence

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.